

WHY CALIBRATE YOUR TEST AND MEASUREMENT EQUIPMENT?

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Modern Test and Measurement Equipment (TME) performs a critical role in providing accurate measurement in commissioning installations and systems or fault finding existing ones. Trades people and technicians place their faith in the accuracy of the measurements performed by this equipment.

As the technology of systems and installations become more complex, the requirement for more sophisticated and accurate test and measurement equipment increases. With high speed processing and communications, fast protection and safety services and state of the art processing and manufacturing systems, contractors and engineering staff need reliable and accurate Test and Measurement Equipment to install, configure, calibrate and fault find correctly.

WHY WOULD A BUSINESS RISK THEIR RESULTS BEING INACCURATE BY NOT HAVING THEIR EQUIPMENT CALIBRATED?

Some important questions to ask yourself...

- Are the results accurate and can they be relied on?
- What would be the impact if you approved incorrect results?
- How important is the accuracy of results you obtain from your test equipment?
- Does your customer expect you to provide evidence of traceable calibration?

The correct operation of Test and Measurement Equipment (TME) is a major contributor to the accuracy and reliability of measurements performed. User training for TME will also improve the quality and reliability of measurement results. Having TME checked at regular intervals will increase the reliability of the measurement results as well as providing traceability. The extent to which formal calibration needs be applied on TME is dependent on the application. For those organisations undertaking work that requires accurate measurements on which the end product quality is dependent, a traceable calibration in accordance with the manufactures procedures should be performed at regular interval.

WHAT IS CALIBRATION?

“The application of specifically known and accurately measured input to ensure that an item will produce a specifically known output which is accurately measured or indicated. Calibration includes adjustment or recording of corrections as appropriate.”

To determine if TME needs calibration these questions need to be asked:

- Identify which measurements are critical to ensuring end product compliance and performance. These measurements may include physical measurements of the product or service (e.g. frequency, voltage or resistance), or values of specified performance parameters (e.g. Bit Error Rate, insertion loss, gain, etc), and process control parameters that are critical to ensuring final product or service quality.
- Identify the tolerance & accuracy required for these measurements.
- Identify the correct test and measurement equipment to perform the task and determine the extent/frequency of calibration required.

SELECTING A CALIBRATION PROVIDER?

Traceability reference instruments/equipment

Traceability of the measurement instrument chain - means that the customer's instrument must be calibrated against a reference instrument that has been calibrated against a more accurate reference instrument and so on until the calibration chain reaches the primary national or international standards. The primary standards in Australia are held by the National Measurement Institute (NMI)

Competence of the Calibration Laboratory

Competence (skill of calibration personnel) -means that the personnel performing the calibration procedure and signing the calibration certificates and test reports have been trained to perform this task and have been assessed as competent at doing so by the equipment manufacturer or a testing authority such as NATA (National Association of Testing Authorities- Australia) A laboratory that is audited and complies with ISO17025 is assessed as competent to produce testing and calibration results and can issue a NATA endorsed test report to the customer.

CALIBRATION INTERVALS

“How often do I need my instrument calibrated?” The answer is not simple and depends on factors such as: the impact of the instrument on quality of the end product, the environment and the way in which the instrument is used and recorded history of the instrument. Ultimately it comes down to the risk period that, should the equipment become faulty, the company is prepared to continue delivering product and services.

Manufacturers often recommend calibration intervals for their instruments models. These recommendations are based on their knowledge of the design of the instrument and the performance history of a number of the same instruments over a period of time.

HOW CAN YOU REDUCE YOUR CALIBRATION COSTS?

- Identify what does and what does not need calibration.
- Obtain advice on whether a traceable calibration should be performed.
- Check with the manufacturer as to the recommended calibration cycle.
- Only purchase equipment that can be calibrated locally (to prevent extra cost in international freight costs and delays)

For more detailed advice as to your specific calibration requirements, please call 1800 256 838